

Weekly Progress Report

EGN4950C Senior Design, Group 22

Kheiven D'Haiti | Jorge Sanchez | Ashleyn Montano | Riley Roberts | Victor De Souza Teixeira

Report Date: September 7, 2026. Period: September 7 to September 13, 2026.

EGN4950C Senior Design

Weekly Progress Report

Project Title: Open Source Selective Video Compression for Static Surveillance Cameras **Technical Sponsor:** Defense Innovation Unit (DIU) / NIWC Pacific, Cody Hayashi **Group:** 22 **Report Date:** September 7, 2026 **Report Period:** September 7, 2026 to September 13, 2026 **Team Members:** Kheiven D'Haiti (CS, AI Minor) | Jorge Sanchez (CS) | Ashleyn Montano (CS) | Riley Roberts (CS) | Victor De Souza Teixeira (CS, Cybersecurity)

NOTE FOR THE TEAM, DELETE THIS BOX BEFORE SUBMITTING. This report is being filed on September 14, one day after the Sunday deadline. Read your own section, correct anything that is wrong, and fill in the two [FILL IN] items in Part 1 before this goes out. Export to PDF named 2026-09-07-Progress-Report-Grp-22.pdf.

Part 1: Team Section

All team members have detailed tasks listed on Teams Planner: YES. Source document: docs/project-records/PLANNER-FALL-2026.md and its companion CSV.

1. Team Meeting

Did the team meet this period? [FILL IN: date, time, and who attended. If the team did not meet, say so and describe how coordination happened instead.]

2. Sponsor / Advisor Meeting

Did the team meet with the sponsor this period? [FILL IN. State when the next sponsor check-in is planned.]

3. Team Progress This Period

This period was dominated by the Module 1 Revised Proposal, due September 13, and by getting the public repository into a state that matches what the team actually intends to ship.

Kheiven completed the two tasks that unblock everyone else's ramp-up (docs/CHANGES-SUMMER-2026.md, and committing the research, security, and project-history documents that had been sitting locally), then spent the rest of the period on a larger repository cleanup that was not on the original plan: rewriting the README on both the desktop and mobile branches for a non-technical reader with an honest

beta section for the Android app, removing AI-development-process documents from the public branch, dropping the dormant commercial-license plan so the project reads as open source only, and archiving the Spring roadmap in favor of a new person-by-person Fall roadmap with a regenerated Teams Planner import. This work was real and needed, but it came at the cost of task 2.3, the system design section of the proposal, which did not get done this period and carries into week 3.

Victor completed his assigned proposal task, 2.7 (organizational chart and completed-tasks list), and opened pull request #15 with the drop-in content. It went through one round of review comments (correcting which milestone the AV1 codec work actually belonged to, and clarifying the two different "Coordinator" titles on the sponsor and student sides) and was merged into `main` on September 14.

Jorge, Ashleyn, and Riley did not complete their assigned proposal sections (2.4, 2.5, 2.6) this period. Each spent the time reviewing the current state of their subsystem instead, and the team is using that review to scope new features to add to the desktop application, rather than only finishing mobile parity work. Kheiven's own focus for the coming period is getting the mobile port working end to end.

The consequence is that the Module 1 Revised Proposal was not fully assembled by the September 13 deadline. Only section 2.7 is ready to fold in; sections 2.3 through 2.6 are outstanding and are the team's first priority this coming week.

4. Challenges and Blockers

- **The Module 1 Revised Proposal deadline was missed.** Only Victor's section (2.7) was completed on schedule. The system design section (2.3), the requirements list (2.4), the literature survey (2.5), and the Gantt chart (2.6) are still outstanding. This report is being filed a day late for the same reason: the week's attention went to repository cleanup instead of the proposal.
- **Three of five members have not yet returned with concrete plans.** Jorge, Ashleyn, and Riley are still reviewing their subsystems and have not yet proposed the specific new desktop features they intend to build. That needs to land early in week 3 so it can inform both the delayed proposal sections and the week 4 task breakdown.
- **The mobile pairing defect and the lack of a mobile test harness remain open,** carried from last period. Building the test harness (tasks 3.1, 3.2) is scheduled first in week 3, ahead of the pairing fix in week 4, for the same reason given last report: diagnosing it without tests is what consumed the most time in August.
- **A documentation accuracy gap was found while rebuilding the roadmap.** `docs/getting-started.md` says `docker compose up --build` works from a fresh clone, but the `Dockerfile` copies in `yolo_v8n.onnx`, which is gitignored and never fetched automatically. Nobody has actually run the Docker install from a clean checkout since that gitignore rule was written. Verifying and fixing this is scheduled as task 3.16.

5. Summary Tables

Previous report period: August 31 to September 6, 2026

Member	Tasks completed	Not completed	Tasks for next period
Kheiven D'Haiti	2	0	3
Jorge Sanchez	2	0	2

Member	Tasks completed	Not completed	Tasks for next period
Ashleyn Montano	2	0	2
Riley Roberts	2	0	2
Victor De Souza Teixeira	2	0	2

Table 1. Previous report summary.

Current report period: September 7 to September 13, 2026

Member	Tasks completed	Not completed	Tasks for next period
Kheiven D'Haiti	2	1	4
Jorge Sanchez	0	1	3
Ashleyn Montano	0	1	3
Riley Roberts	0	1	3
Victor De Souza Teixeira	1	0	4

Table 2. Current report summary. "Not completed" for Jorge, Ashleyn, and Riley is their proposal section (2.4, 2.5, 2.6 respectively), carried into week 3 alongside their originally scheduled tasks.

Part 2: Individual Sections

Each member writes their own section. Task numbers match the MS Teams Planner and docs/project-records/PLANNER-FALL-2026.md.

Kheiven D'Haiti, CS Major, AI Minor

Report Date: September 7, 2026

Tasks completed this reporting period

Task 2.1: Write the summer changes document for the team (completed)

Description: The team needed one document that explains what changed over the summer and why, rather than a 158-commit log nobody has time to read.

Outcome: docs/CHANGES-SUMMER-2026.md written and committed, covering the distribution work, the compression quality changes, the security audit, and the mobile app build-out, with a reading guide pointing each member at the files relevant to their subsystem.

Task 2.2: Commit the stranded documentation and re-verify push status (completed)

Description: docs/RESEARCH.md, docs/SECURITY.md, and docs/archive/PROJECT-HISTORY.md existed only on my machine as of last report.

Outcome: All three committed and pushed. Re-checked every branch against its remote; zero unpushed commits remained after this task.

Additional work this period (not on the original plan)

The rest of the period went into a repository cleanup pass that the proposal work made necessary once I looked closely at what a sponsor or outside contributor would actually see on the public branch:

- Rewrote the README on both main (desktop first, then mobile) and mobile/app (mobile first) for a non-technical reader, with an honest beta section for the Android app covering what it does, what it does not do yet, and the real risks (pairing not surviving a restart, uploads not surviving the app being killed, thin test coverage, no signed build yet).
- Removed AI-development-process documents (planning and handoff notes, AGENTS.md) from the public main branch, since a sponsor or outside contributor has no use for the internal record of how the code was developed. These stay on mobile/app, the internal working branches.
- Dropped CLA.md and LICENSE-COMMERCIAL.md, the dormant dual-license plan, and rewrote every "premium tier" reference in code comments and docs to plain "optional extras" language, so the project reads as open source only. A commercial variant, if it happens, will be a separate forked repository.
- Archived the Spring 2026 roadmap to docs/archive/ROADMAP-SPRING-2026.md and wrote a new ROADMAP.md for this semester, organized by person with a linked section each, detailed for weeks 2 through 6 and outlined for weeks 7 through 14 to match the planner's own rolling window.
- Regenerated the Teams Planner CSV to include the week 3 application-testing tasks that had been added to the written plan but never made it into the importable file, plus a new task (3.16) to verify the Docker install path, which I found undocumented while doing this work.

Outcome: The public branch now matches what the team is actually building and intends to ship, and the semester plan is current. The cost was task 2.3, below.

Tasks not completed this reporting period

Task 2.3: Revised proposal, system design section and functional diagrams Not started. The architecture changed enough over the summer (mobile client, device token boundary, push path) that this is closer to a rewrite than an edit, and the cleanup work above took priority. Carried into week 3.

Planned tasks for the coming period (September 14 to September 20)

Task 2.3 (carried over): Revised proposal, system design section and functional diagrams. Redraw the architecture diagram to include the mobile client, the device token boundary, and the push path, marked in red.

Task 3.1: Make `SvcApi` fakeable and add JVM unit tests for the view models. gradlew testDebugUnitTest should cover the pairing, library, events, and home view models.

Task 3.2: Add a debug build variant with HTTP logging for diagnosis, so a failing mobile request shows its reason in logcat instead of being a mystery on a minified build.

Task 3.11: Fresh-install walkthrough, with the rest of the team: install `SVCS-Setup.exe`, complete first-run Setup, run one compression, all without opening the source code.

Task 3.16: Verify the Docker install path end to end on a fresh clone, and fix `docs/getting-started.md` to either state the `yolov8n.onnx` prerequisite plainly or make the build fetch it automatically.

Jorge Sanchez, CS Major

Report Date: September 7, 2026

Tasks completed this reporting period

None formally closed. I spent the period reviewing the ingest and streaming code I own (the watchfolder daemon and `MultiFrameSource`) against what changed over the summer, and I am scoping new capture and multi-camera streaming features to bring to the team for the desktop app, rather than building yet. Task 2.4, the requirements section of the revised proposal, was not completed this period.

Planned tasks for the coming period (September 14 to September 20)

- Finish scoping new desktop ingest and streaming features and bring specific proposals to the team.
- **Task 2.4 (carried over):** Revised proposal, requirements list including mobile requirements, marked in red.
- **Task 3.3:** Investigate the `WorkManager` API and draft the upload worker design.
- **Task 3.4:** Audit the resumable upload protocol for resume correctness.
- **Task 3.11:** Fresh-install walkthrough, with the rest of the team.

Ashleyn Montano, CS Major

Report Date: September 7, 2026

Tasks completed this reporting period

None formally closed. I spent the period reviewing the metadata and query code I own against what changed over the summer, and I am scoping new event surfacing and search features to bring to the team for the desktop app. Task 2.5, the literature survey section of the revised proposal, was not completed this period.

Planned tasks for the coming period (September 14 to September 20)

- Finish scoping new desktop event and search features and bring specific proposals to the team.
- **Task 2.5 (carried over):** Revised proposal, literature survey and reference list.
- **Task 3.5:** Build the desktop `EVENTS` panel, read-only table.

- **Task 3.6:** Add the empty state and ten second auto-refresh to the EVENTS panel.
 - **Task 3.13:** Add a folder-browse button to the Setup destination field.
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Riley Roberts, CS Major

Report Date: September 7, 2026

Tasks completed this reporting period

None formally closed. I spent the period reviewing the compression mode code I own against what changed over the summer, and I am scoping new compression and zone-related features to bring to the team for the desktop app. Task 2.6, the Gantt chart section of the revised proposal, was not completed this period.

Planned tasks for the coming period (September 14 to September 20)

- Finish scoping new compression and zone features and bring specific proposals to the team.
 - **Task 2.6 (carried over):** Revised proposal, Gantt chart built from the planner.
 - **Task 3.7:** Add the GET /api/zones/frame route returning a still frame.
 - **Task 3.8:** Draft the zone editor canvas layout.
 - **Task 3.12:** Test the Compact install path with no FFmpeg on PATH.
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Victor De Souza Teixeira, CS Major, Cybersecurity Minor

Report Date: September 7, 2026

Tasks completed this reporting period

Task 2.7: Revised proposal, organizational chart and completed-tasks list (completed September 12, merged September 14)

Description: The assignment requires an organizational chart that explicitly names a team leader, and a table of tasks completed to date. I drafted both as drop-in content for Kheiven to fold into the master proposal document, opened as pull request #15 (feature/victor-week2-task-2.7).

Implementation: The organizational chart names Kheiven D'Haiti as Student Team Leader / Project Coordinator, distinct from Geena Wann-Kung's "Project Coordinator" title on the sponsor side, with the sponsor and faculty advisor above him and the four subsystem owners reporting to him, matching the ownership table in ROADMAP.md. The completed-tasks table (proposal Table 4.4) is split into three periods: Spring 2026 milestones 1 through 3 (complete) with the open items honestly marked as not done by the May 6 deadline, Summer 2026's headline build-out, and Fall Week 1's ten completed ramp-up tasks.

After opening the pull request I addressed two review comments: the AV1 codec work had been miscredited to the Spring capstone when it was actually finished afterward in a summer fix, and I moved it to the correct period; and the two different "Coordinator" titles needed an explicit note that there is no reporting relationship between them.

Outcome: docs/project-records/PROPOSAL-SECTION-2.7-ORG-CHART-AND-COMPLETED-TASKS.md, merged into main in pull request #15.

Evidence: Pull request #15, commits b92fd02 and 4c7b18c.

Planned tasks for the coming period (September 14 to September 20)

This is my heaviest week this semester, four tasks against everyone else's two or three:

Task 3.9: Read `push_notify.is_safe_push_url` and specify the webhook emitter, reusing the existing guard.

Task 3.10: Write the socket-server test harness for webhook delivery, in the style of `tests/test_push_notify.py`.

Task 3.14: External network penetration test against a running SVCS instance, findings recorded against `docs/SECURITY.md`.

Task 3.15: Fuzz the video-ingest and upload path with malformed media.

If this proves too much alongside the others, tasks 3.9 and 3.10 move to week 4 rather than rushing the security testing.